

HW03

ENAE 601: Astrodynamics

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Dr. Healy

Vai Srivastava

Problem 1**Curtis 4.3**

For a geocentric satellite whose inertial position and velocity vectors in a geocentric equatorial frame are:

$$\mathbf{r} = 2500\hat{\mathbf{I}} + 16000\hat{\mathbf{J}} + 4000\hat{\mathbf{K}} \quad (\text{km})$$

$$\mathbf{v} = -3\hat{\mathbf{I}} \quad -\hat{\mathbf{J}} + 5\hat{\mathbf{K}} \quad (\text{km s}^{-1})$$

- a) Find the classical orbital elements e , a , i , Ω , ω , θ .
- b) Compute the equinoctial orbital elements a , h (*this is not angular momentum*), k , p (*this is not semilatus rectum*), q , λ .

a) Answer

b) Answer

Problem 2**Curtis 4.4**

At a given instant, the position $\mathbf{r}$ and velocity $\mathbf{v}$ of a satellite in the geocentric equatorial frame are:

$$\mathbf{r} = -13000\hat{\mathbf{K}} \quad (\text{km})$$

$$\mathbf{v} = 4\hat{\mathbf{I}} + 3\hat{\mathbf{J}} - 3\hat{\mathbf{K}} \quad (\text{km s}^{-1})$$

- a) Find the classical orbital elements e , a , i , Ω , ω , θ .
- b) Compute the equinoctial orbital elements a , h (*this is not angular momentum*), k , p (*this is not semilatus rectum*), q , λ .

a) Answer

b) Answer

Problem 3**Curtis 4.5**

At time t_0 (relative to perigee passage) the position $\mathbf{r}$ and velocity $\mathbf{v}$ of a satellite in the geocentric equatorial frame are:

$$\mathbf{r} = 6500\hat{\mathbf{I}} - 7500\hat{\mathbf{J}} - 2500\hat{\mathbf{K}} \quad (\text{km})$$

$$\mathbf{v} = 4\hat{\mathbf{I}} \quad + 3\hat{\mathbf{J}} - 3\hat{\mathbf{K}} \quad (\text{km s}^{-1})$$

- a) Find the classical orbital elements e , a , i , Ω , ω , θ .
- b) Compute the equinoctial orbital elements a , h (*this is not angular momentum*), k , p (*this is not semilatus rectum*), q , λ .

a) Answer

b) Answer

Problem 4**Curtis 4.6**

With respect to the geocentric equatorial frame, the position vector of a spacecraft is $\mathbf{r} = -6000\hat{\mathbf{I}} - 1000\hat{\mathbf{J}} - 5000\hat{\mathbf{K}}$ (km) and the orbit's eccentricity vector is $\mathbf{e} = 0.4\hat{\mathbf{I}} + 0.5\hat{\mathbf{J}} + 0.6\hat{\mathbf{K}}$. Calculate the true anomaly θ if the satellite is approaching perigee.

a) Answer

Problem 5**Curtis 4.7**

Given that, relative to the geocentric equatorial frame, $\mathbf{r} = -6600\hat{\mathbf{I}} - 1300\hat{\mathbf{J}} - 5200\hat{\mathbf{K}}$ (km), the eccentricity vector is $\mathbf{e} = -0.4\hat{\mathbf{I}} - 0.5\hat{\mathbf{J}} - 0.6\hat{\mathbf{K}}$, and the satellite is flying toward perigee, calculate the inclination of the orbit.

a) Answer

Problem 6

Find the equinoctial elements for the following orbits:

a) Position and Velocity:

$$\mathbf{r} = \begin{bmatrix} 3698.3455 \\ -34232.4473 \\ 0 \end{bmatrix} \text{ (km)}, \quad \mathbf{v} = \begin{bmatrix} 2.1482 \\ -1.4944 \\ 0 \end{bmatrix} \text{ (km s}^{-1}\text{)}$$

b) Position and Velocity:

$$\mathbf{r} = \begin{bmatrix} -21082.0848 \\ 36515.242 \\ 0 \end{bmatrix} \text{ (km)}, \quad \mathbf{v} = \begin{bmatrix} -2.6627 \\ -1.5373 \\ 0 \end{bmatrix} \text{ (km s}^{-1}\text{)}$$

c) Position and Velocity:

$$\mathbf{r} = \begin{bmatrix} -15843.4562 \\ -2247.7766 \\ 21200.4627 \end{bmatrix} \text{ (km)}, \quad \mathbf{v} = \begin{bmatrix} 1.4544 \\ -3.5188 \\ 0.7138 \end{bmatrix} \text{ (km s}^{-1}\text{)}$$

a) Answer

b) Answer

c) Answer

Code

```
./src/index.py Python
1  import numpy as np
2  import matplotlib.pyplot as plt
3  import os
4
5
6  def heading(problem):
7      print(f"\n-----\n {problem}\n-----")
8
9
10 def main():
11     #####
12     # p02 #
13     #####
14     heading("p02")
15
16     x = np.linspace(0, 10, 100)
17     y = np.sin(x)
18
19     print(f"x:\n{x}\nny:\n{y}")
20     with open("./outputs/text/s02.txt", "w", encoding="utf-8") as f:
21         f.write(f"x:\n{x}\nny:\n{y}")
22
23     plt.figure()
24     plt.plot(x, y)
25
26     save_path = os.path.join(os.path.dirname(__file__), "./outputs/figures/s02.png")
27     plt.savefig(save_path)
28     plt.show()
29
30
31 if __name__ == "__main__":
32     main()
```